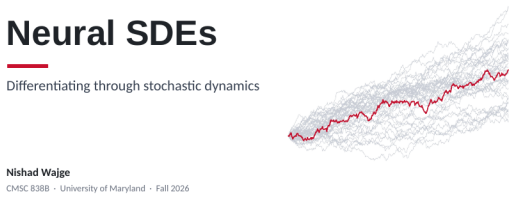
Neural SDEs: Practice Script

25:00 slot = 20:00 talk + 5:00 discussion. Black = say it. Gray = only if asked. Practice with a timer and check the timestamps.

Slide 1 · 0:00 – 0:20



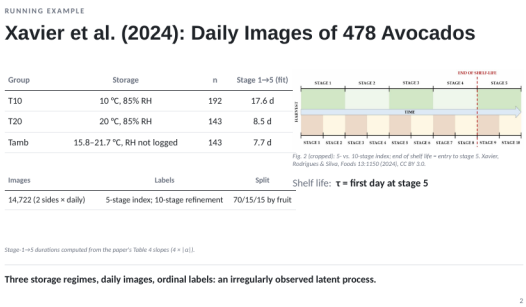
PLAN: 25:00 total = 20:00 talk + 5:00 discussion. If behind at slide 7 or 19, skip them.

CHECKPOINTS: slide 9 by 6:00 · slide 14 by 12:15 · slide 20 by 18:30 · discussion by 20:00.

Appendix (A1–A10) is not presented; jump there only if asked.

- Hi, I'm Nishad. Today: neural SDEs, and how to take gradients through randomness.
- One running example the whole way: when does an avocado go bad?

Slide 2 · 0:20 – 1:10

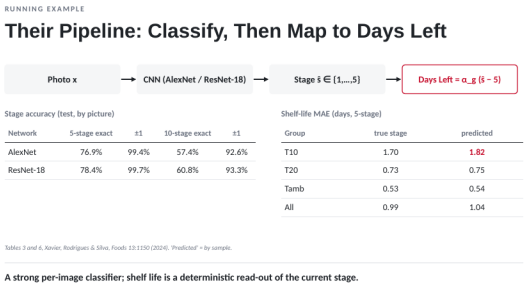


- Data: Xavier, Rodrigues and Silva, Foods 2024, open access. 478 Hass avocados photographed daily on both sides until overripe.
- Three groups: 10 °C, 20 °C (both 85% humidity), and room temperature, which drifted between 15.8 and 21.7 °C with humidity not logged.
- Labels: a 5-stage ripening index, refined to 10 stages. End of shelf life is reaching stage 5 (stage 9 on the 10-stage scale).
- Last column: from their fitted slopes, going from stage 1 to 5 takes ~17.6 days at 10 °C, ~8.5 at 20 °C.
- Splits are by fruit, so no avocado appears in both train and test.

IF ASKED — why is the 10-stage index there? Finer resolution; the paper reports it's less accurate exactly (≈60%) but ≈93% within one stage.

IF ASKED — license? The paper is CC BY 3.0; figures are reused with attribution. The dataset is on Mendeley Data (doi:10.17632/3xd9n945v8).

Slide 3 · 1:10 – 2:00

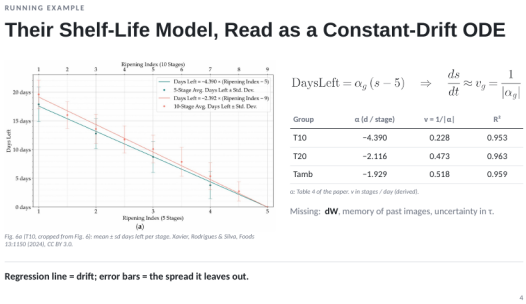


- Their method: a CNN with transfer learning classifies the stage from one photo; a per-group linear regression turns the stage into days left.
- Left: ~78% exact on 5 stages, >99% within one stage. 10 stages: ~60% exact, ~93% within one.
- Right: mean absolute error in shelf life. Even with the TRUE stage label ('true stage' column), error is 1.7 days at 10 °C. So most of the error isn't the classifier; it's the stage → days map.

IF ASKED — the abstract says 88.8%? That's accuracy on the best side of each fruit; the table shows per-picture averages.

IF ASKED — what's 'by sample'? Each fruit has two photos (sides); 'by sample' uses the better/combined side, which improves accuracy because one side can have marks.

Slide 4 · 2:00 – 2:50

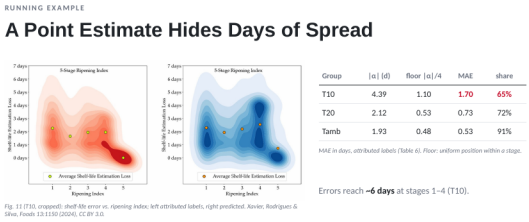


- Their linear regression, Days Left = α(stage − 5), can be read as a constant-velocity ODE in stage: ds/dt ≈ v with v = 1/|α|.
- Rates: 0.23 stages/day at 10 °C, 0.47 at 20 °C, 0.52 at room temperature. 20 °C ripens ~2× faster than 10 °C, as the paper reports.
- Left plot (their Fig. 6a, T10): the error bars are standard deviations of days left at each stage, often ±2–3 days. A deterministic ODE can't represent that spread; a diffusion term can.

IF ASKED — is that inversion exact? No: regressing days on stage and inverting isn't the same as regressing stage on days (slopes differ by a factor R², ~4% here); stage is ordinal; and their fit is pinned at stage 5. Close enough to read off rates.

IF ASKED — R² ≈ 0.96 looks great; why change it? R² measures fit of the mean; the decision (sell/discount by Friday) needs the distribution around the mean, which R² doesn't capture.

Slide 5 · 2:50 – 3:45



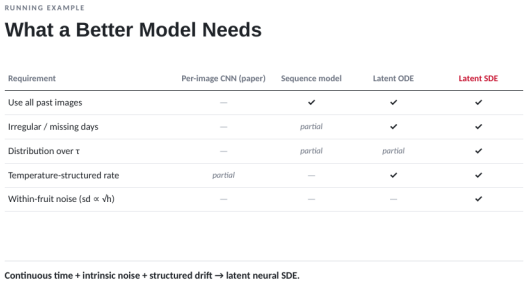
- Fig. 11, 10 °C group: shelf-life error by stage. Left human labels, right model predictions.
- Average error ≈ 2 days at stages 1–4, tails to ~6 days.
- Part is unavoidable: one label can't tell day 1 of a stage from day 4. That floor, |α|/4, is 65% of the error at 10 °C, 91% at room temperature.
- The rest is what a model must represent as a distribution. It's largest at 10 °C, the longest horizon.

IF ASKED — couldn't this just be label noise? Partly: the 'attributed' labels are human. Separating diffusion (sd ∝ √horizon) from fruit-to-fruit rate variation (sd ∝ horizon) needs the per-stage sds in Fig. 6.

IF ASKED — why |α|/4? The MAE of a uniform distribution of width w about its center is w/4.

IF ASKED — why zero at stage 5? Days left is 0 by definition once the fruit is in stage 5.

Slide 6 · 3:45 – 4:20

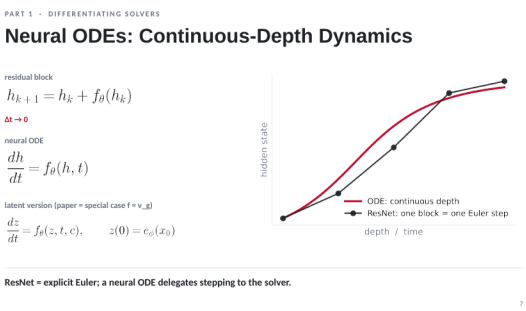


- The paper's model uses one image and outputs a point. Rows: what we want.
- A sequence model gets partway. A latent ODE handles irregular time, but has no noise that keeps arriving.
- Latent SDE: all five by construction; its diffusion adds variance that grows like √h along the way.

IF ASKED — can't a latent ODE spread? Yes, via an uncertain initial state or per-fruit rate (sd ∝ horizon), but that collapses as images accumulate.

IF ASKED — 'partial' for the paper on temperature: they fit a separate slope per storage group; that's temperature-dependent but not a function of temperature (can't interpolate to 15 °C).

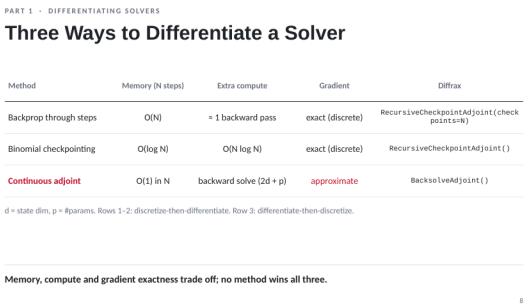
Slide 7 · 4:20 – 5:00 — SKIP-IF-SHORT (most of the room knows this).



- Residual block h + f(h) = one explicit Euler step. Δt → 0 gives dh/dt = f_θ(h, t); the solver chooses the steps.
- Bottom: in the avocado setting, z(0) is an encoding of the first photo, f evolves it under storage condition c. The paper's model is the special case f = constant rate v_g in stage space.

IF ASKED — costs? Each forward pass is a solve; adaptive solvers' step counts (NFE) tend to grow during training.

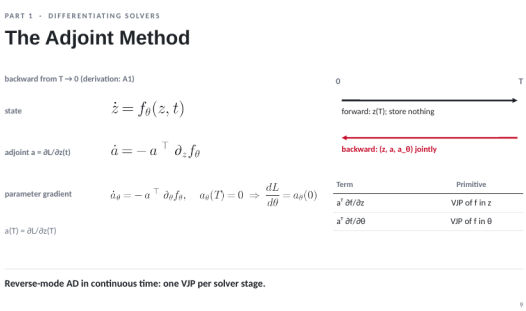
Slide 8 · 5:00 – 5:55



- Row 1: treat the solver as code and backprop. Exact gradient of the computed solution; stores every step.
- Row 2: Griewank's binomial checkpointing (revolve): O(log N) checkpoints, O(N log N) recomputation, still exact.
- Row 3: derive the continuous gradient as a backward ODE (next slide) and solve it. Memory independent of N, but the backward re-solve doesn't retrace the forward discrete path, so the gradient is approximate.

IF ASKED — 2d + p? Backward state = z (d), adjoint a (d), running parameter gradient (p).

Slide 9 · 5:55 – 7:25 — core slide.

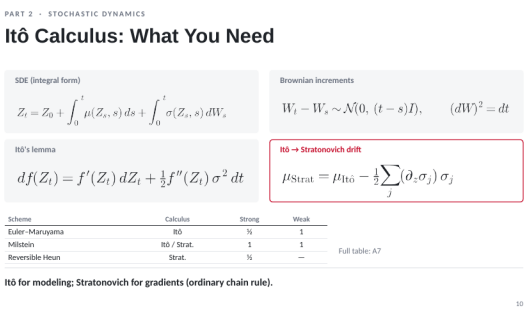


- a(t) = ∂L/∂z(t): sensitivity of the loss to the state at time t. Linear ODE backward from a(T) = ∂L/∂z(T).
- a_θ starts at 0 at T; its value at 0 is dL/dθ.
- z is re-integrated backward rather than stored: source of O(1) memory and of the approximation.
- Both terms are vector–Jacobian products: one reverse-mode call of f per solver stage (Lecture 4's primitive). Never form the Jacobian.

IF ASKED — derivation? Appendix A1 (Lagrange multiplier on ż = f, integrate by parts).

IF ASKED — relation to Pontryagin? a is the costate; the adjoint ODE is the costate equation.

Slide 10 · 7:25 – 8:55

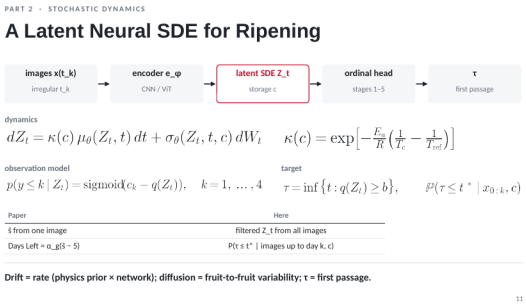


- An SDE is shorthand for the integral equation; the dW integral is Itô (left endpoint).
- Increments are independent N(0, Δt); (dW)² = dt is why stochastic calculus differs.
- Itô's lemma: the extra ½ f″ σ² dt term. Example: d log S picks up −½σ².
- Stratonovich (midpoint) obeys the ordinary chain rule; the drift changes by −½ Σ_j (∂σ_j/∂z) σ_j. Zero when σ doesn't depend on z.
- Orders: Euler–Maruyama strong ½ (paths), weak 1 (expectations).

IF ASKED — strong vs weak? Strong: E|Z_T − Ẑ_T|. Weak: |E f(Z_T) − E f(Ẑ_T)|. Probabilities like P(τ ≤ t) need weak; pathwise gradients care about strong.*

IF ASKED — which does Euler converge to? Euler–Maruyama → Itô; Heun / midpoint → Stratonovich.

Slide 11 · 8:55 – 10:15

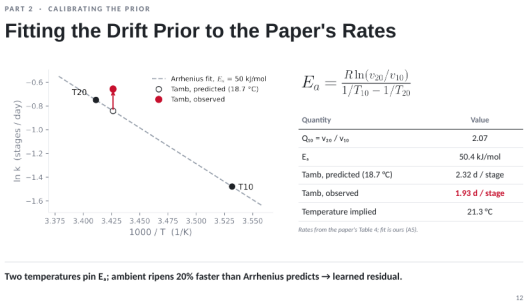


- Encode each photo; Z_t evolves as an SDE between photos; an ordinal head maps Z to stage probabilities; shelf life τ is when the ripeness score crosses the stage-4/5 cut.
- Drift: Arrhenius factor κ(c) times a neural rate. E_a activation energy, R gas constant, T in kelvin. Diffusion σ_θ: variability across fruit and storage.
- Ordinal head: cumulative-logit model, P(y ≤ k) = sigmoid(c_k − q(Z)), cut points c_k ordered. b = c_4 defines 'end of shelf life'.
- Bottom: what changes vs the paper. They classify one image and map to a point; we filter all images and output a probability.

IF ASKED — how do you compute P(τ ≤ t)? Monte Carlo from the filtered posterior at the last photo; count crossings, with a Brownian-bridge correction between steps (A6).*

IF ASKED — should σ scale with temperature? If temperature speeds up the whole biological clock, σ ∝ √κ. Slide 13 tests exactly this against the paper's numbers.

Slide 12 · 10:15 – 11:10

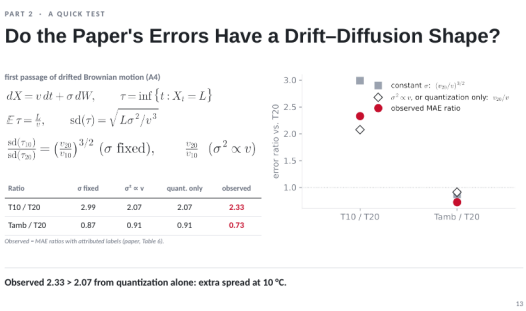


- Using their per-group slopes: rates 0.228 (10 °C) and 0.473 (20 °C) stages/day. Q10 ≈ 2.07, so E_a ≈ 50 kJ/mol, a plausible enzymatic value.
- Predict room temperature (mean 18.7 °C): 2.32 days per stage. Observed: 1.93. Ambient fruit ripened ~20% faster than temperature alone explains; you'd need 21.3 °C, near the top of the recorded range.
- Likely culprits: humidity wasn't controlled in Tamb, and temperature fluctuated. Point: a physics prior gets you most of the way; a learned residual absorbs what it misses. That's why the drift is κ(c) × network.

IF ASKED — is two points enough to fit Arrhenius? It determines E_a exactly but gives no check of the functional form; that's why I treat it as a prior, not a law.

IF ASKED — could temperature fluctuation explain it? Rate is convex in T, but with sd 1.2 °C the Jensen effect is under 1%. Not enough for 20%.

Slide 13 · 11:10 – 12:15



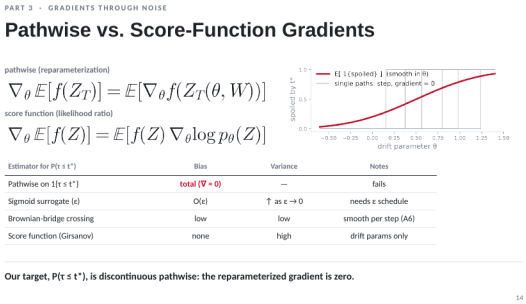
- Back-of-envelope: stage drifts at rate v with noise σ. Time to cover distance L is inverse Gaussian: mean L/v, sd $\sqrt{(L\sigma^2/v^3)}$.
- If σ is the same at all temperatures, the T10/T20 error ratio is $(v_{20}/v_{10})^{3/2} \approx 3.0$. If temperature speeds up the whole clock ($\sigma^2 \propto v$), it's $v_{20}/v_{10} \approx 2.1$.
- Careful: quantization alone also gives 2.07.
- Observed: 2.33. Above quantization, so extra spread at 10 °C. Three numbers can't pin σ(T); it's a shape check, not a fit.

IF ASKED — Tamb is lower than every prediction? Consistent with Tamb being a different regime (uncontrolled humidity, fluctuating temperature), as slide 12 showed.

IF ASKED — why MAE ∝ sd? For a fixed distribution shape, MAE is a constant multiple of sd (0.8 for Gaussian). That assumes small CV; under fixed σ the inverse-Gaussian CV differs 1.44× between T10 and T20, so treat the ratios as rough.

IF ASKED — caveats? MAE of a stage-based estimate, not the sd of τ, and labels are human. Derivation in A4.

Slide 14 · 12:15 – 13:45

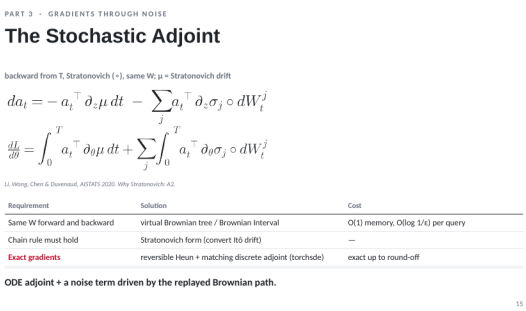


- Pathwise: Z_T as a deterministic function of θ and fixed noise W; differentiate inside E. VAE reparameterization trick. Low variance; needs f differentiable.
- Score function: E[f ∇ log p_θ]; any f, high variance.
- Plot: for each fixed noise path, 'spoiled by t*' is a step in θ, derivative zero almost everywhere. The expectation (red) has nonzero slope. Pathwise returns 0: a biased gradient, not a noisy one.
- Table: four estimators. Brownian-bridge crossing probability is the one I'd use: smooth in the endpoints, low variance, exact for frozen coefficients within a step.

IF ASKED — Girsanov ratio? For drift parameters: ∇_θ log p = ∫ (∂_θ(σ⁻¹μ))ᵀ dW. Parameters in σ change quadratic variation → path measures mutually singular in continuous time; on a grid the EM density exists but its estimator's variance blows up as Δt → 0.

IF ASKED — finance analogue? Digital and barrier options: same problem; same fixes (smoothing, likelihood-ratio Greeks, bridge corrections). Mohamed et al. JMLR 2020 for the ML view.

Slide 15 · 13:45 – 15:05



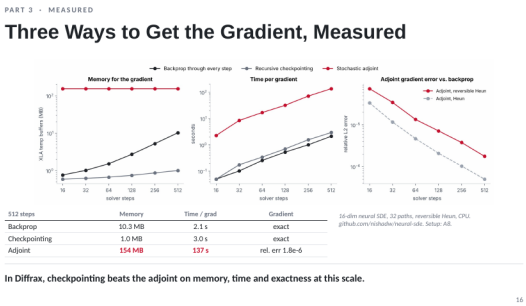
- Compare slide 9: one extra term per noise channel, $a^\top \partial\sigma_j/\partial z \circ dW^j$.
- Requirement 1: the backward pass must see the same noise. A virtual Brownian tree regenerates W(s) from a seed (A6).
- Requirement 2: Stratonovich form, so the ordinary chain rule holds. Convert an Itô model's drift first, or the gradients are wrong.
- Exact version: reversible Heun with its matching discrete adjoint (Kidger 2021, torchsde). Diffrax's BacksolveAdjoint is approximate.

IF ASKED — conversion formula? $\tilde{\mu} = \mu - \tfrac{1}{2} \Sigma_j (\partial\sigma_j/\partial z)\, \sigma_j$. Zero when σ doesn't depend on z.

IF ASKED — why is reversible Heun exact? It's algebraically reversible, so the backward pass recovers the forward states to round-off; pair it with the discrete adjoint of that scheme.

IF ASKED — Itô directly? Possible, but the backward-in-time Itô SDE needs extra correction terms.

Slide 16 · 15:05 – 15:45

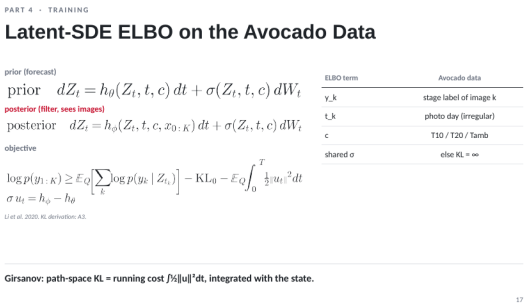


DEMO: no time in the talk. If someone asks during discussion, play assets/demo.mp4 (52 s).

- Memory (log scale): backprop linear in steps, checkpointing ~log, adjoint flat but ~150 MB. The method needs only 2d + p ≈ 110 KB; the rest is a Diffrax/XLA implementation constant growing with params and noise channels (A8).
- Time: adjoint ~60× slower.
- Error: adjoint vs backprop falls ~1/steps: discretization error of optimize-then-discretize. Plain Heun slightly better than reversible Heun here: reversibility alone doesn't make the continuous adjoint exact.
- Matches Diffrax's own recommendation: use checkpointing.

IF ASKED — why so large? Most likely per-noise-channel parameter cotangents materialized (512 × 12.7k × 8 B ≈ 52 MB, plus temporaries); measured scaling is consistent, not profiled buffer by buffer.

Slide 17 · 15:45 – 17:05



- Latent SDE (Li et al. 2020): a VAE whose latent is a path.
- Prior: our ripening SDE given only storage. Posterior: same σ, drift also sees the images (e.g., a GRU run backward over encoded photos gives a context per time).
- ELBO: expected log-likelihood of observed stage labels at photo times − KL of initial state − KL between the SDEs.
- Girsanov: path KL = E $\int$ ½‖u‖² dt with σu = h_φ − h_θ. Append it as one extra state; each solve gives a Monte Carlo estimate.
- Gradients of all of it flow through the solver: slides 8–16.

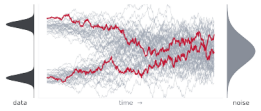
IF ASKED — why shared σ? Girsanov changes drift only; different $\sigma\sigma^\top$ ⇒ different quadratic variation ⇒ mutually singular measures ⇒ KL = ∞.

IF ASKED — forecasting? After the last photo, switch to the prior and sample forward for $P(\tau \leq t^*)$.

Slide 18 · 17:05 – 18:00

PART 4 · CONNECTION

Diffusion Models Are Neural SDEs



forward (fixed)

$$dx = f(x,t)\,dt + g(t)\,dW$$

reverse (learned score)

$$dx = \left[f(x,t) - g(t)^2\, s_\theta(x,t)\right]dt + g(t)\,dW$$

probability-flow ODE

$$\frac{dx}{dt} = f(x,t) - \tfrac{1}{2}\,g(t)^2\, s_\theta(x,t)$$

denoising score matching

$$\min_\theta \; \mathbb{E}_{t,x_0,\varepsilon}\left\|s_\theta(x_t,t) - \nabla_{x_t}\log p_t(x_t \mid x_0)\right\|^2$$

	Latent SDE (avocado)	Diffusion model
learns	μ, σ	score only
training	backprop through solver	simulation-free

Diffusion = neural SDE with fixed σ, trained simulation-free; ours learns σ and backprops through the solver.

18

- Song et al. 2021: a fixed forward SDE turns data into noise. Reversing it needs the score ∇ log p_t; learn it as s_θ. The reverse SDE is a neural SDE.
- Probability-flow ODE: same marginals, deterministic sampling.
- Training is simulation-free: p(x_t | x_0) is Gaussian in closed form. No solver, no adjoint.
- Contrast: our avocado model learns the dynamics, so it must backprop through the solve.

IF ASKED — who showed the reversal? Anderson (1982).

IF ASKED — flow matching? Trains the probability-flow ODE's velocity directly; π0 robot policies use it.

IF ASKED — likelihoods? Change of variables along the ODE, with a Hutchinson trace estimator.

Slide 19 · 18:00 – 18:30 — SKIP-IF-SHORT.

BEYOND PRODUCE

Same Structure Elsewhere

Domain	Latent state	Neural SDE role	Reference
Weather	atmosphere	diffusion sampling of ensembles	Price et al., Nature 2025 (GenCast)
Robotics	action trajectory	reverse diffusion / flow matching	Chi et al. 2023; Black et al. 2024
Quant finance	price, volatility	calibrated market models	Gierjatowicz 2022; Cohen 2021
Clinical	patient state	irregular vitals (ODE/CDE family)	Rubanova 2019; Kidger 2020
Generative models	image / audio	score-based sampling	Song et al. 2021

Wherever decisions depend on the distribution over trajectories.

19

- Same pattern as the avocado: hidden state, randomness, sparse observations, decisions on the distribution of futures.
- My repo also has the finance case: a neural local-vol SDE calibrated to a Heston smile by backprop through Monte Carlo (0.31 vol pts RMSE vs 4.21 flat).

Slide 20 · 18:30 – 19:30 — analysis portion.

LIMITS

Open Problems

Issue	Symptom	Evidence / mitigation
Adjoint error	backward re-solve ≠ forward path	ANODE (2019); 1e-4–1e-6 here
Stiffness	NFE grows during training	kinetic / Jacobian regularization (Finlay 2020)
Discontinuous targets	∇ 1[τ ≤ t*] = 0 pathwise	bridge crossing, surrogate, score function
Identifiability	drift ↔ diffusion trade-off; latent reparam.	ordinal head anchors q; synthetic recovery test
Visual confounds	lighting, marks, wind rub look like ripening	paper: per-sample > per-picture accuracy
Uncontrolled covariates	Tamb 20% faster than Arrhenius	condition on humidity if logged

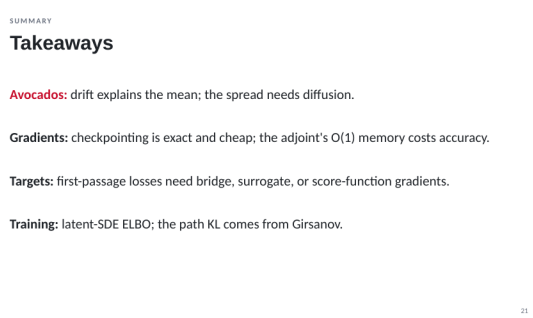
O(1) memory is real; exact, unbiased, identifiable gradients each cost extra.

20

- Adjoint: not the gradient of what the forward solver computed. ANODE showed it can be badly wrong; 1e-4 to 1e-6 here.
- Stiffness: learned dynamics stiffen and solvers slow down; regularize kinetic energy or Jacobian norms.
- Our target is discontinuous: naive pathwise training gives zero gradient.
- Identifiability: fast drift + small noise can mimic slow drift + big noise. Test on synthetic data with a known SDE.
- Paper-specific: surface marks look like ripening, and ambient storage doesn't follow temperature.

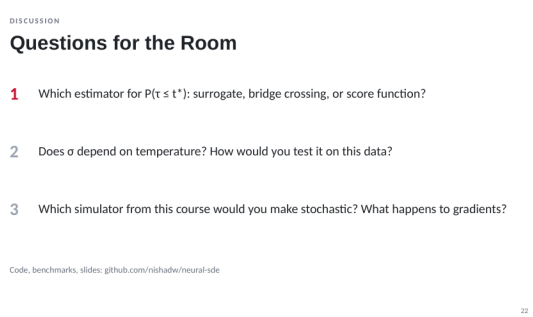
IF ASKED — latent reparameterization? Any invertible map of the latent space leaves the likelihood unchanged; the ordinal head anchors the ripeness direction.

Slide 21 · 19:30 – 20:00



- Four things to remember. Read each line once; don't add new material.
- Then: 'Let's open it up. Three questions to start.'

Slide 22 · 20:00 – 25:00 — hard stop at 25:00.



~1:30 per question, then close.

- Q1: bias vs variance. Bridge crossing: exact per step for frozen coefficients, low variance. Score function: unbiased, noisy, drift params only. Sigmoid: needs ε annealing.
- Q2: fit the latent SDE with $\sigma_\theta(c)$ free per group, compare to $\sigma \propto \sqrt{\kappa(c)}$ by held-out likelihood; or estimate per-group variance of daily stage increments directly from the label sequences.
- Q3 (Prof. Lin's lectures): stochastic forcing adds Monte Carlo variance and needs same-noise replay; for contact-rich systems noise can smooth the loss landscape (randomized smoothing).
- Close at 25:00: 'Thanks. Code and slides are on GitHub.' Appendix is for questions only.